SPE-230166-MS

Innovative Approach to Managing High Wax Content in an Un-Piggable Deepwater Subsea Gas Pipeline

Y. F. Youssef, Pharaonic Petroleum Company, Cairo, Egypt

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Atoll field is a high-pressure high-temperature subsea gas field located in the East Mediterranean at 930-meter water depth. The field is producing from the Oligocene layer, Contains a high condensate gas ratio of 29 STB/MMSCF with high wax content and WAT of 37°C. The objective is to manage wax deposition issues in this Atoll production system and hydrate formation in the 111 km pipeline.

Wax deposition is managed by injecting continuous wax inhibitor through the umbilical, which decreases the WAT to around 18°C. A two-stage process for wax mitigation is proposed. The first stage involves a chemical soak using xylene and hexane. The second stage includes a pigging campaign. The team proposed a virtual pigging system using MEG injected from onshore to the well flowlines through the subsea umbilical, in addition to shut down wells to allow liquid redistribution in the pipeline.

The wax precipitation leads to a production decrease of 1.0 MMSCFD without any change in well performance, meaning the field would reach zero production in less than one year without intervention. Once the wax precipitates at a certain distance (first 20 km section), it behaves like an orifice, causing additional JT cooling effect and leading to hydrate formation. This created a complex and challenging problem of wax and hydrate formation in the subsea pipeline system. Atoll pipeline experienced three full blockage events. The virtual pigging job was repeated many times, resulting in increased field production compared to the condition before the intervention. PHPC established field-operating guidelines that defined intervention triggers, outlined methodologies, and updated the hydrate management strategy to account for the JT cooling effect caused by wax buildup. These improvements significantly reduced the need for interventions from four times a year to potentially just once annually. This innovation delivered a total gain of 76.5 K MMSCFD of gas and over 1.86 million bbl of condensate between 2021 and 2024 compared to 2020.

This pioneering innovation of virtual pigging using MEG through the umbilical in a subsea pipeline without pigging facilities offers a new approach to wax and hydrate management. The methodology provides valuable insights for similar subsea developments and contributes to improving flow assurance under extreme offshore conditions

Introduction

In March 2015, BP Egypt announced Atoll a new gas discovery in the North Damietta Offshore Concession in the East Nile Delta. Atoll is a recent Oligocene (085) discovery in the East Nile Delta (END). It lies in deep water (- 950m) and deeper, higher pressure & temperature reservoirs north of the current Taurt &Ha'py producing fields. The current reference case is based on a 3-well (Phase 1) early production scheme (EPS) with a potential (Phase 2) involving further 4 wells full field development which would sanctioned in immediately after Phase 1. The Phase 1 development case envisages a new subsea system at Atoll tied back to the existing Taurt export line to the West Harbor onshore facility.

As shown below in Figure 1, Atoll gas will be produced via a new sharp inclined ~43km 20in production pipeline from Atoll Manifold to the Taurt South field. At Taurt South, Atoll production will be tied into the existing ~68km 20in export pipeline from Taurt South to the onshore West Harbor gas processing facilities. This routing will be designated as the high-pressure system.

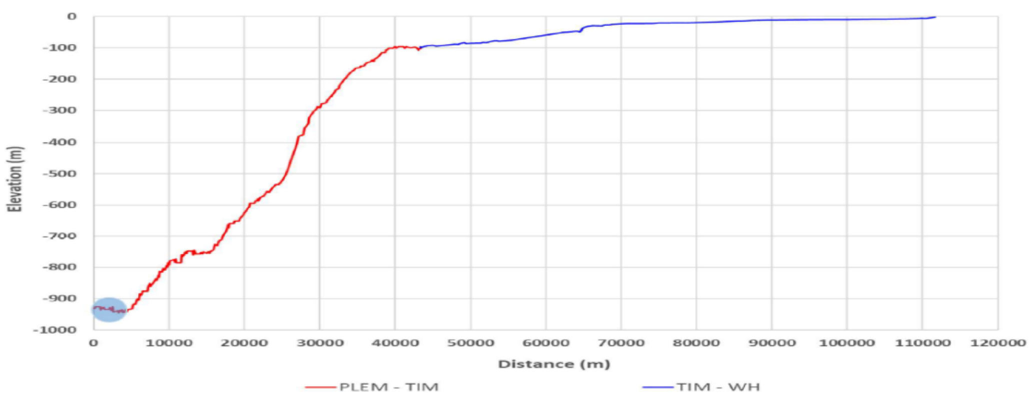

Figure 1—Atoll PL Profile.

Following the successful start-up of Atoll EPS, the company completed a fourth well, which began delivering production in October 2020. Production from this fourth Atoll well was tied back to an available slot on the Atoll manifold. Figure 2 shows the overall schematic of the development

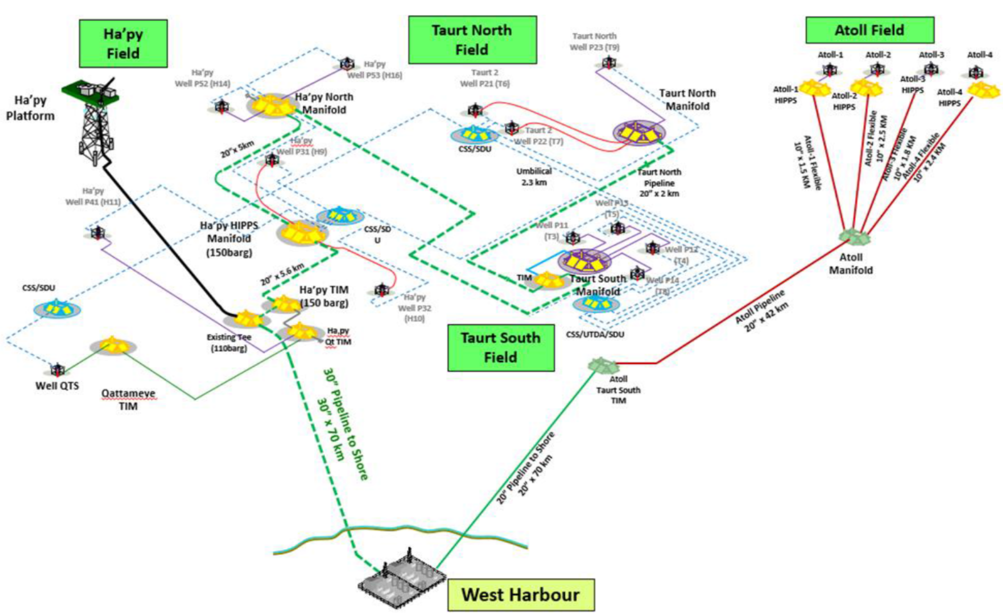

Figure 2—Atoll Subsea Schematic

Problem Statement

Flow assurance is the practice of maintaining a safe, steady, and efficient flow of hydrocarbons from the reservoir to processing facilities or the sales point. It focuses on detecting and managing risks that may reduce or block flow, especially the formation of solid deposits like wax and hydrates. In deep-water gas-condensate developments such as the Atoll field, these challenges become more severe due to cold seabed temperatures and long subsea pipeline distances.

Atoll's primary flow assurance challenges are wax deposition, hydrate plugging, and liquid hold-up. These issues are affected by temperature and pressure. For example, when liquids accumulate, it can increase the chance of hydrates formation, and both wax and hydrates can even form together under specific circumstances. A variety of strategies is needed to overcome these problems.

Wax deposition is a major flow assurance challenge in deep-water oil and gas production, particularly in fields such as Atoll, where wax buildup in subsea oil pipelines arises when the operating temperature drops below the Wax Appearance Temperature (WAT), commonly referred to as the cloud point. At these conditions, Wax can lead to two concerns. First, Precipitated wax in bulk fluids causes viscosity increases and complex rheological behavior. Second, cold fluids and cold internal pipe wall surfaces can cause deposition of wax, which increases frictional pressure drops, could in the extreme lead to pipeline and Flowline blockages and can cause operational problems in its removal or for inspection. (Harris 2016)

To manage wax effectively, a combined approach is commonly employed:-

- Chemical Injection, into the flow stream to prevent or reduce wax crystal formation and deposition.
- Thermal insulation, Applying insulation to pipelines to maintain fluid temperatures above WAT, thereby preventing the wax from solidifying and depositing.
- Pigging. Running mechanical pig through the pipeline to scrape and remove wax deposits that have accumulated on the inner walls. (N.Haines 2017)

As shown in Figure 3 The volume of condensate obtained from all of the Atoll samples was minimal, this has shown that for representative flow cases, high rates of deposition of paraffin wax can be expected in the early part of the Flowline, typically within 20 km of the pipeline. Accumulation of the deposited wax will inevitably lead to increased pressure drops and consequent production impacts.

Figure 3—Atoll-1 stock tank condensate sample (MPSR-BA-2991)

Two Atoll condensate samples were tested and a further pressurized sample was flashed and tested for WAT and WDT (Harris, 2016). Results are shown in Table 1

Table 1—WAX & WDT temperatures

Sample	WAT (°C)	WDT (100% melt) (°C)
Atoll sample MPSR-BA-2991	34	40
Atoll sample MPSR-BA-2512	32	42
Pressure bar	**WAT °C**	**WDT**
117.5	37	45
65	38	47
1	38	54

Results showed that Atoll condensate has a high Wax Appearance Temperature (WAT) of 37 °C, while the ambient seabed temperature is significantly lower at approximately 13 °C. Although a wax inhibitor is injected to reduce WAT to around 18 °C, the temperature of the produced fluids still drops below this threshold during subsea transport, triggering wax precipitation and deposition. Significant wax deposition is expected in the 20" pipeline within the first 20–30 km, before the fluid temperature reaches the ambient 13 °C.

Table 2 below illustrates the pour point and gelation temperatures respectively that are obtained following two standard pre-treatments for Atoll stock tank condensate they are:

1. The IP Upper method involves heating crude oil to 48°C and then allowing it to cool. However, this temperature is often too low to fully redissolve all the wax that has precipitated out of the crude. This can lead to inconsistent and non-repeatable viscosity data, often showing a higher viscosity than the crude's true state due to the remaining wax crystals.
2. In contrast, the Beneficiated method heats the crude oil to a higher temperature of 80°C before it is cooled at a controlled, static rate. This higher heating temperature ensures that all precipitated wax fully redissolves, resulting in reliable and reproducible viscosity data. Typically, this method reveals a lower, more accurate viscosity for the crude oil.

Table 2—Pour point & Gel point

Sample	IP15	Beneficiated
	Pour Point °C	
Atoll sample MPSR-BA-2991 Depth 5773.8m	**27**	**27**
Atoll sample MPSR-BA-2512 Depth 5736m.	**21**	**21**
	Gelation point °C	
Atoll sample MPSR-BA-2991 Depth 5773.8m	24	24
Atoll sample MPSR-BA-2512 Depth 5736m.	20	20

This wax accumulation results in increased pressure drop, flow restriction, and potential equipment fouling—and in severe cases, may cause complete pipeline blockage, posing high operational and financial risks.

The lack of fluid samples and the short timeline for the project represent major challenges to the selection of appropriate chemicals for the duties outlined in section above. It had been anticipated to utilize fluid from West Akhen field as an analogue to Atoll for testing purposes. The West Akhen field in the Nile Delta produces gas condensate from a HPHT reservoir. The fluid is waxy with a wax content of 13.9 Wt%, a stabilized fluid WAT of 39 °C and a pour point of 22–25°C (Chapman 2016)

Hydrate formation in subsea pipelines becomes more severe when combined with wax buildup, liquid holdup, and challenging pipeline geometry. In the Atoll pipeline, the first section of the 43 km tieback is sharply inclined (See Figure 1). When pressure and temperature drop into the hydrate formation zone—often due to the Joule-Thomson JT cooling effect across areas restricted by wax—hydrates can form quickly. Wax deposits reduce the pipeline's internal diameter and act as insulation, allowing fluid near the pipe wall to cool even further. This creates ideal conditions for hydrates to grow. Additionally, during low flow conditions, liquid holdup can occur, where water and condensate accumulate in low points of the pipeline. These stagnant pockets further increase the risk of hydrate formation. As hydrates begin to form in wax-covered sections with liquid holdup, they can block more of the flow path and combine with wax to form a thick, sticky mixture. This hydrate-wax slurry is highly viscous, increases pressure drop, restricts flow, and can lead to complete pipeline blockage.

The Atoll hydrate dissociation curve is shown in Figure 4. The curve was generated using Multiflash v.6.0 using the CPA EOS. (PhPC 2016)

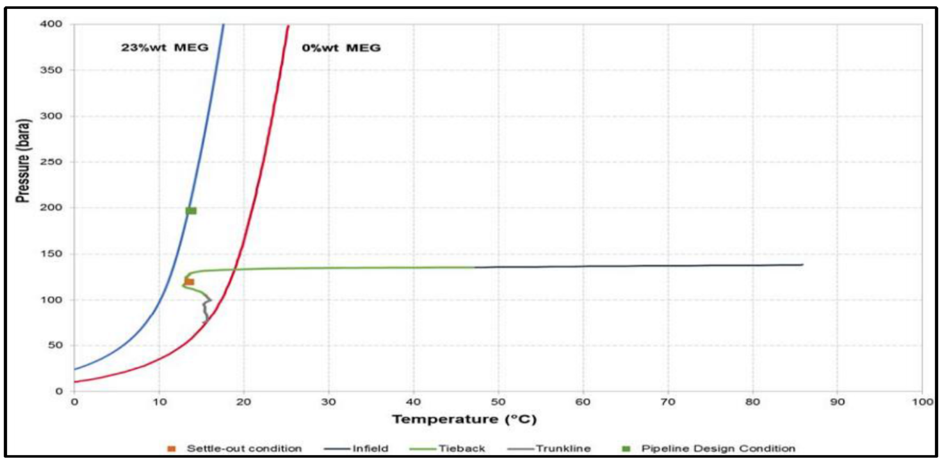

Figure 4—Normal Operating Curve with Hydrate Curves (Design Rates/Early LoF)

The gas velocity and the pipeline's physical layout typically influence liquid holdup in a pipeline. However, the formation of wax and hydrates, especially within steeply inclined sections, can severely restrict the flow of liquids. This leads to significant fluid buildup in these areas, posing a third critical flow assurance challenge.

Atoll pipeline history

After successful startup of Atoll field started up in December 2017, The PhPC operations team have reported gradual increases in pressure drop over the Atoll pipeline system during the curse of 2018. Attempts to maintain throughput have been undertaken by increasing flow and hence pressure downstream of the HIPPS; this has resulted in occasional shutdowns to the system through incurring high-pressure alarms. Additionally, reduced arrival pressures have been applied. Which reviewed the pressure profile data and modelled the potential ID reduction to match the existing process conditions undertook a preliminary assessment. This assessment concluded there was a risk of wax deposition occurring in the first 20 km of the Atoll pipeline.

The Atoll pipeline experienced three full blockage events, the first in February 2020 during a ramp-up of well production from 250 MMSCFD (obligated due to low gas demand) to full rate. This triggered the high differential pressure trip set point, causing a sudden shutdown of the wells. The initial hydrate management strategy assumed a MEG concentration of 23% by weight would suffice to inhibit hydrate formation, with a 2–3 °C temperature suppression margin under critical operating conditions as shown in

Figure 4. However, the immediate cause of the blockage was believed to be hydrate formation, triggered by prior wax deposition. Wax accumulation created flow restrictions that acted like orifices, intensifying the local Joule-Thomson (JT) cooling effect and shifting conditions into the hydrate formation region.

Post-event analysis confirmed that the existing MEG concentration was inadequate for the evolving thermal and hydraulic conditions in the pipeline. Some wax was recovered at the onshore facility, indicating partial displacement, and this contributed to improved pipeline performance. Consequently, a revised MEG injection strategy was implemented, targeting concentrations above 30% by weight to provide a greater safety margin against hydrate formation.

Figure 5 shows the second blockage occurred on July 2020, when the Atoll pipeline experienced a sudden pressure increase that triggered a 10-minute shutdown of the wells. This was caused by severe slug flow, or liquid off-loading, within the steep 43 km tieback. At the time, the system was operating close to the XT trip limit. The likely cause was the buildup of viscous liquids in the tieback during extended low-flow periods (150–180 MMSCFD), which increased the risk of wax accumulation. The steep slope of the 43 km subsea line further complicated the situation.

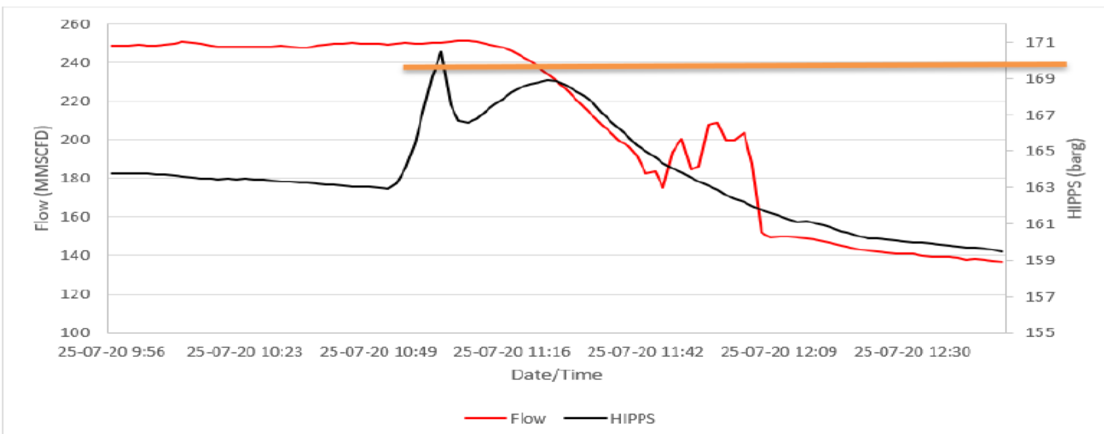

Figure 5—Second Blockage Event

Refer to Figure 6, which illustrates the third blockage incident in the Atoll pipeline, during a period when the pipeline was maintaining stable production rates exceeding 300 MMSCFD. HIPPS pressure was close to the XT trip limit. At the same time, excess differential pressure (DP) increased from 35 to 53 barg, suggesting gradual flow restriction due to wax or hydrate buildup. The third blockage occurred following delayed intervention, during a restart attempt using reverse to start-up wells from nearby fields after depressurization, the buildup caused a blockage.

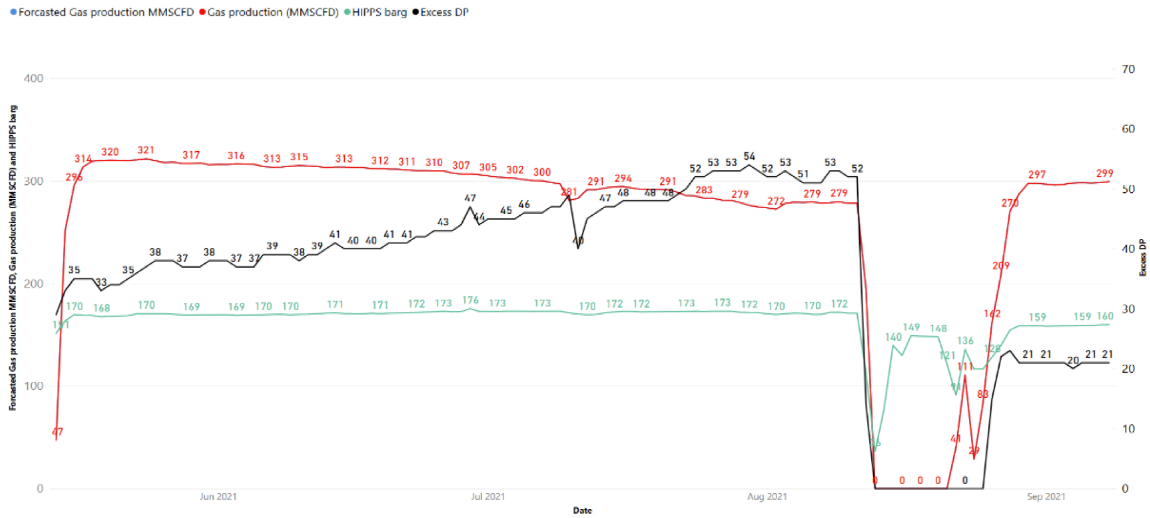

Figure 6—Third Blockage Event

Wax Modelling

Developed flow assurance wax model utilizing OLGA software, but with wide range of assumption and uncertainties. March 2019-Schlumberger utilized OLGA Wax Deposition Model; wax layer could deposit in first 20 Km of tie back only no wax deposition remaining in pipeline. See Figure 7 (Billdal 2019)

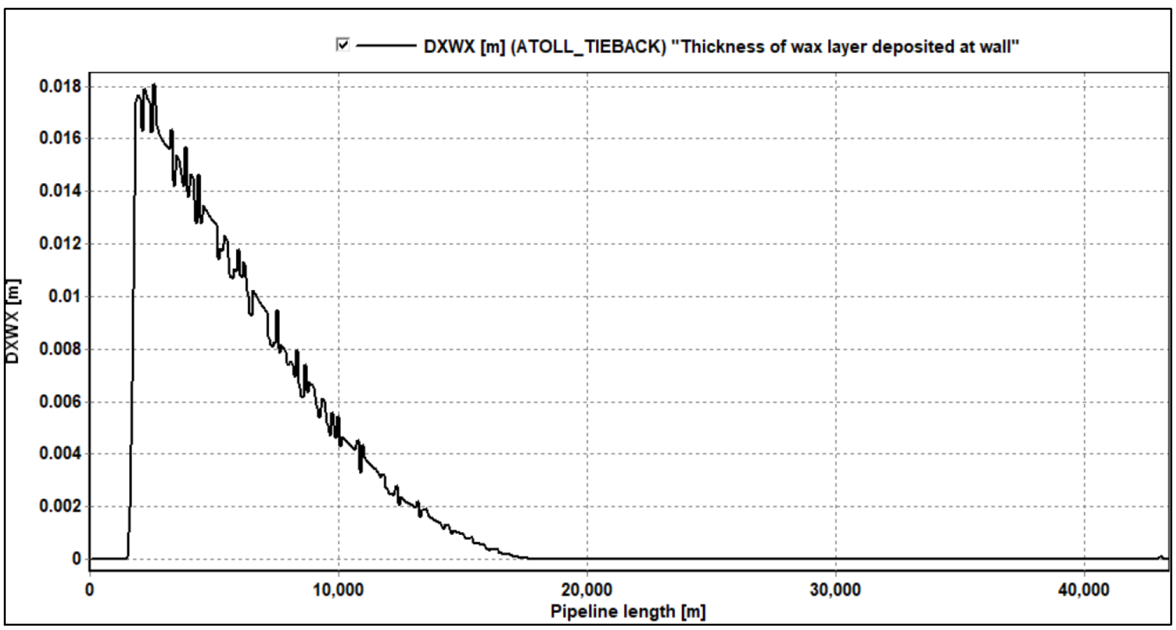

Figure 7—Wax deposition modelling of Atoll Tieback 20 km

November 2019-BP Flow assurance UEC performed tuning for OLGA wax model and stated, that Wax could extend from Atoll PLEM until + 40km; soaking activity should give us some useful feedback on wax extent (See Figure 8).

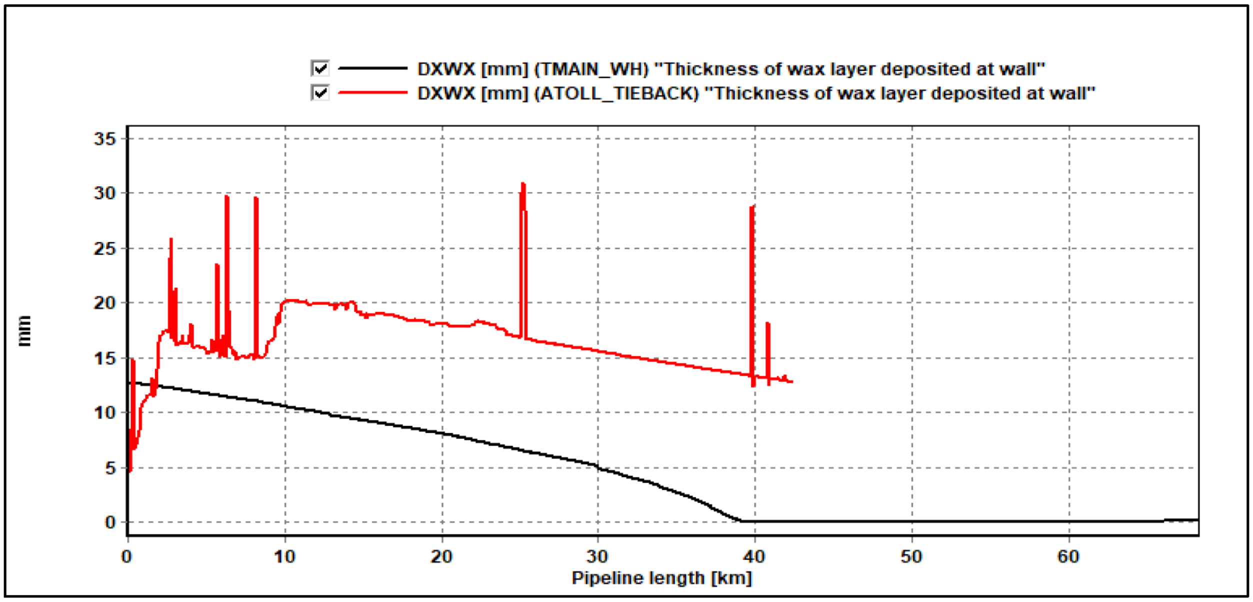

Figure 8—Wax deposition modelling of Atoll Tieback 40 km

May 2020-BP Flow assurance UEC build another model to explain the reason for performance improvement from November 2019 until May 2020 as shown in Figure 9. Model assumed two scenarios that wax was removed form steep section, another scenario that removed from shallow section. It could not be definitively determined what happened in practice, nor the precise location from which the wax was removed.

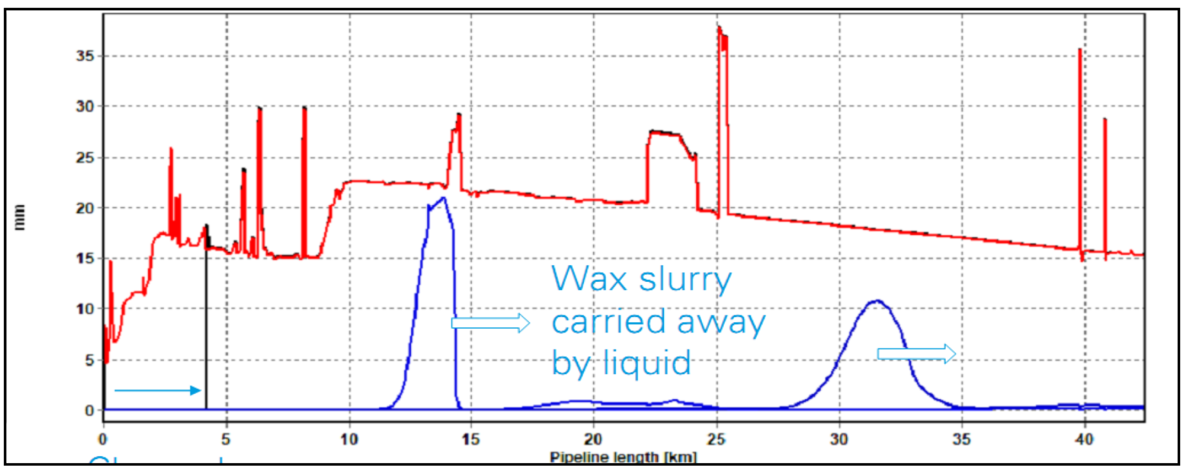

Figure 9—Improvement modelling of wax removal

Wax Mitigation Options for Atoll

During a preliminary discussion, four particular options were discussed with reference to the Atoll system. These include strategies based on avoiding deposition, whether by passive insulation or active heating; limiting the impact of the deposition through use of chemical inhibitors; and, using solvents, mechanical pigging.

1. Insulate the pipeline to keep fluid temperatures above WAT. However, this was not considered feasible for the long distance pipeline

2. Deploy chemical Inhibitors to help with wax management. It was noted, however, that they could not completely eliminate deposition in the 20" pipeline.
3. Mechanical Pigging
4. Do nothing. This would allow wax to be deposited in the 20" pipeline over the life of the early production scheme. However based on the prediction results it is unlikely that this would be feasible, with the large quantities of wax that would be deposited, leading to reduction in ID and possible blockage, as well as corrosion issues

Atoll Wax Recovery Programme; a two stage process is proposed. The first stage is a preliminary chemical soak and assessment. If the soak has been successful in returning the pressures and flow rates then the pipeline will be placed back into production without further intervention (other than development of a long-term wax mitigation plan). If the process conditions cannot be restored a second phase of the wax clearance plan will be implemented which will concentrate on a progressive pigging campaign. (PhPC 2018)

A temporary remedial solution was adopted by injection of around 70 cubic meters of Xylene as a wax solvent. The wells would be shut off and the xylene would be injected through the spare chemical core in the umbilical then a soaking time of 20 hours was allowed before restarting the wells. This solution has showed slight success in recovering some of the excess pressure drop across the pipeline a preliminary study was done. Which estimated 550-600 m3 of wax deposit in the pipeline.

Figure 10 shows the initial and revised ID has to achieve the constant volume required: Based on the expected deposition for the first 20km, it was checked what ID reduction for the first 20 km of the 20″ Flowline would be required for a constant flowrate to give this increase in pressure across the system. The calculation is necessarily simplified and assumes an equal 3-leg supply of 115 MMSCFD into the 20″ pipeline and 345 MMSCFD. The thickness of deposit on the wall (if assumed constant over the 20km) will be 21mm approximately. This calculates to be a total volume of some 555m3 of solids.

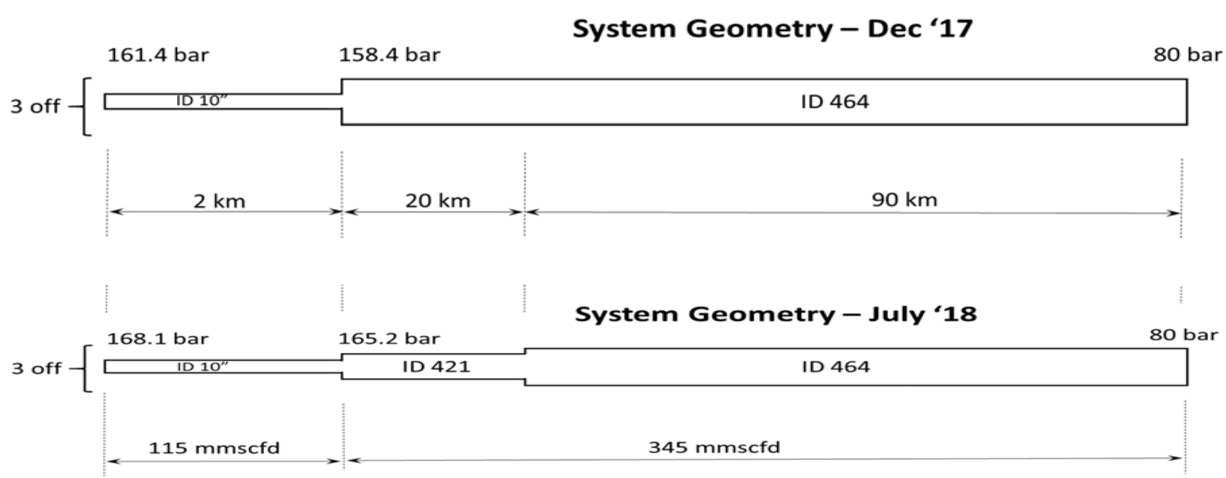

Figure 10—ID Reduction to Simulate Wax Deposition

Mechanical Pigging

Mechanical pigging over such a long pipeline with multiphase fluids is technically complex and risky. The biggest challenge is successfully sending a pig through the line without it being stuck. This is tough because we do not know the exact internal shape of the pipeline due to wax deposition, in addition to a sharp inclined 43 km section.

Prior to running pigs it is necessary to assess as accurately as possible mechanical properties of the wax: its strength and how much there is, especially its thickness around the pipe. Controlling the pig's pressure and maintaining a steady pig speed also present significant challenges. All these factors dramatically increase the risk of the pig being stuck, which could lead to a complete wax blockage (a "wax candle") in the pipeline causing significant gas deferral (lost production and revenue). However, due to unpredictable pig performance and how many runs it will take to clean the line. Beyond the immediate losses, there is a critical risk of losing the pipeline for a prolonged period, or even permanently. Consequently, it was concluded that the pipeline is unsuitable for pigging.

Chemical Soaking

In this technique, approximately 20,000 bbls of solvent were injected into tieback from onshore following pipeline depressurization. The wells were then restarted to sweep the solvent through the system, aiming to dissolve and mobilize wax deposits. However, this option was eliminated due to the following reasons: The solvent soaking process faced several challenges. First, its effectiveness was uncertain due to a lack of detailed information about the exact location, thickness, and distribution of wax deposits along the pipeline. Additionally, the operation necessitated a full pipeline depressurization, resulting in an extended shutdown period that adversely affected the production. Moreover, the procedure involved injecting approximately 20,000 barrels of solvent, introducing significant operational complexity and safety risks. This required meticulous planning, careful chemical handling, and tight coordination across multiple systems to mitigate potential environmental and safety incidents. (Centre 2019)

Innovative solutions

Dynamic Hydrate Control

Atoll fluids are produced within the hydrate formation zone. To prevent hydrate formation, a Monoethylene Glycol (MEG) concentration of 23% by weight is required assuming the full internal cross-section of the pipeline is available for flow.

In practice, however, pipeline operation is highly dynamic. The extent of wax deposition on the pipeline walls varies over time, causing localized flow restrictions. These restrictions lead to additional cooling via JT effect, potentially resulting in temperatures lower than the expected minimum ambient temperature. To dynamically manage the hydrate formation risk, PhPC has developed an in-house Excel-based tool that estimates temperature drops in the pipeline caused by expansion cooling due to wax-induced flow constriction. The tool calculates the temperature reduction corresponding to various excess differential pressures (DP) and determines the required MEG concentration to effectively inhibit hydrate formation.

Note: Excess DP = Actual Pressure Drop – Clean Pipeline Pressure Drop. For example, at a flow rate of 300 MMSCFD, if the actual DP is 90 bar and the clean pipeline DP is 70 bar, the excess DP due to wax is 20 bar.

This methodology has been validated against real field data on multiple occasions; however, once the excess DP exceeds 20 bar, pipeline operation becomes significantly more challenging, as temperature continues to drop, potentially forming a high-viscosity slurry of wax and hydrates that severely restricts flow.

Virtual Pigging

MEG, while primarily known for inhibiting hydrates, also serves a vital role in wax remediation within pipelines due to its high density and the principle of fluid dynamic shear. This process referred to virtual pigging as shown in Figure 11. This is achieved by first shutting in all wells and then filling the entire Flowline with a MEG plug. Its movement through the pipeline creates a scrubbing action at the pipe wall. Subsequently, by ramping up the wells at a higher rate, the increased fluid flow provides additional force, intensifying this cleaning effect. While not the direct scraping action of a mechanical pig, this concentrated volume of flowing fluid and the subsequent high-rate flow can dislodge and sweep away wax deposits once they are softened or made less adhesive by the MEG, effectively clearing them from the pipe wall and achieving the desired wax removal.

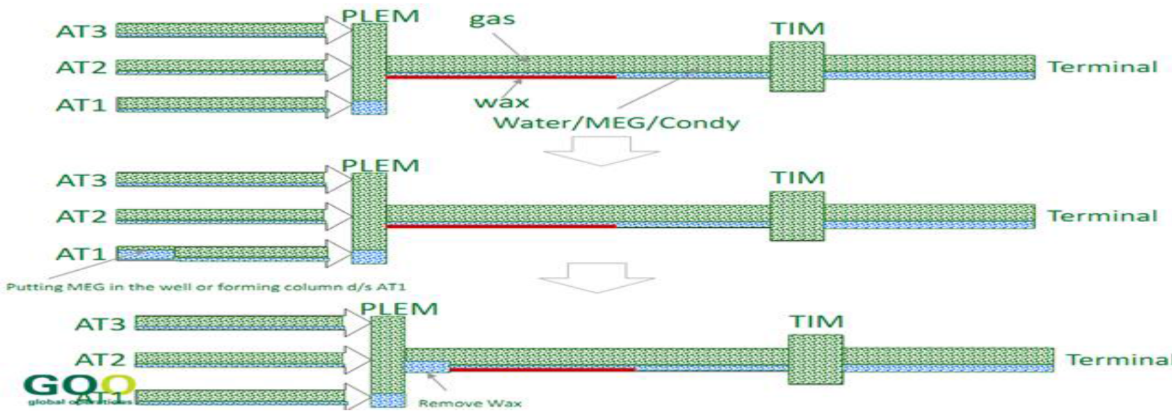

Figure 11—Virtual Pigging Concept

Continuous improvements to this methodology have aimed at maximizing operational efficiency. This has been achieved by lowering the suction pressure through the compressor. These enhancements have significantly reduced production deferrals from 900 MMSCFD per intervention to 370 MMSCFD, and have decreased the intervention frequency from four times per year to only once per year."

Furthermore, Flow assurance team utilized methanol injection 6 hours weekly for the past year as experienced that Methanol decrease the density of production fluids decreasing normal operation liquid hold-up and transporting wax content from the pipeline to the onshore terminal with no deferrals at all.

Figure 12 shows that production has consistently risen since 2020. Starting at 90.7K MMSCFD, it generally increased each year, reaching 116.6K MMSCFD by 2024. The slight dip in 2023 was due to Turnaround (TAR) activity, not a deterioration in performance. Overall, production has gained 76.4K MMSCFD since 2020

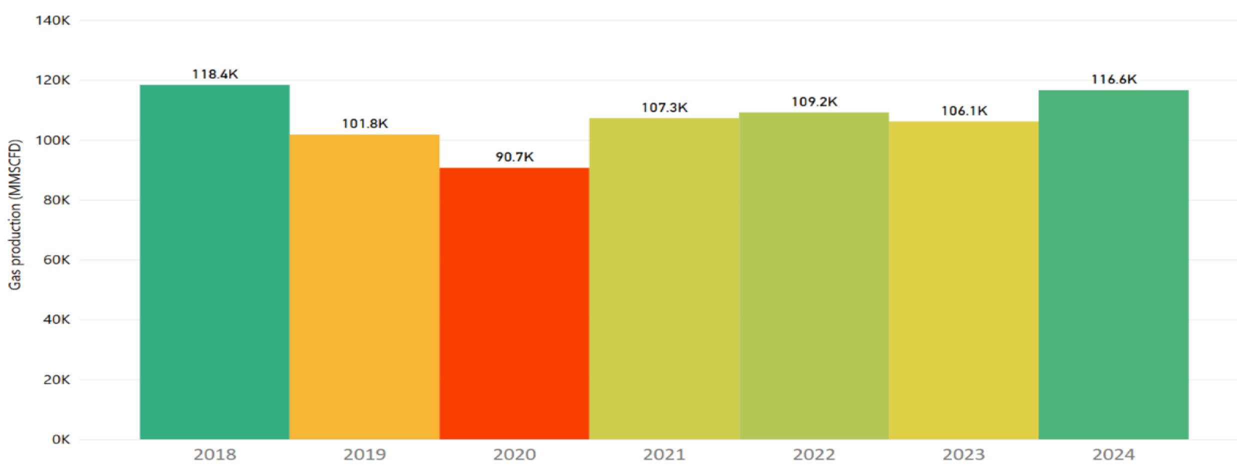

Figure 12—Atoll Gas Production

Figure 13 shows also condensate production has also shown steady growth, rising from 2.95 million barrels in 2020 to 3.49 million barrels in 2024, marking a total increase of 1.85 million barrels over the same period.

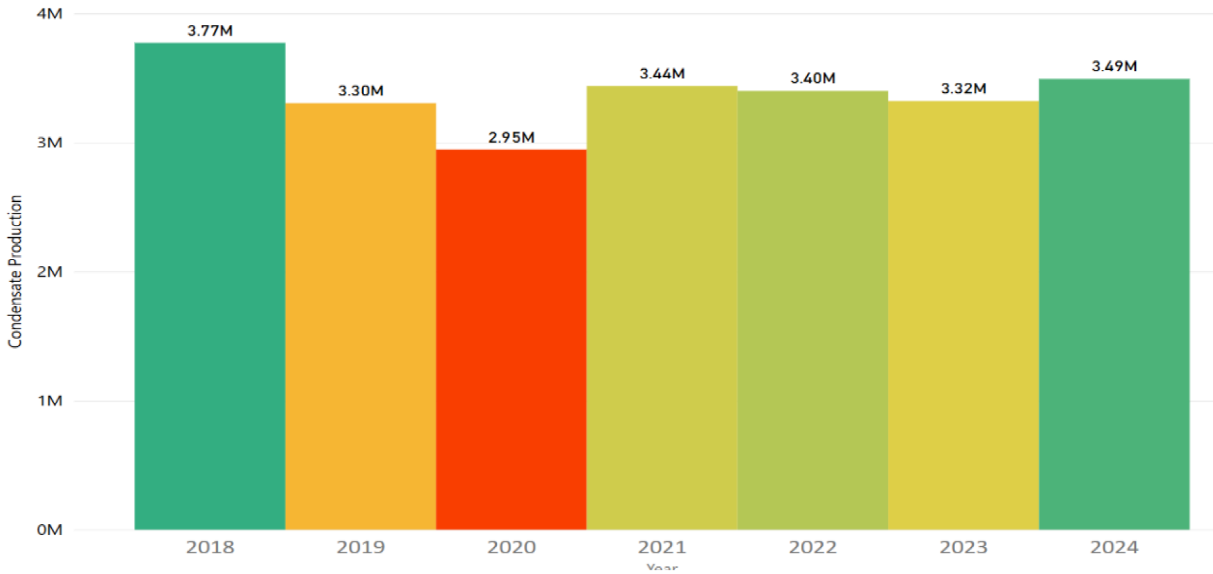

Figure 13—Atoll Condensate Production

PHPC has enhanced its pipeline performance through the development of a pioneering virtual pigging technique. Building on the initial success, the company conducted 17 additional interventions. The Figure 14 illustrates the Atoll pipeline's operational performance. The red curve represents the gas production rate, the green curve shows the HIPPS pressure, and the black curve shows the excess differential pressure caused by wax buildup. The gradual increase in excess differential pressure indicated a deterioration in pipeline performance, prompting the need for intervention to restore optimal conditions. Following the most recent intervention, the pipeline demonstrated improved performance, characterized by a higher gas production rate and reduced excess differential pressure—indicating that the system is operating efficiently and no longer requires frequent annual interventions.

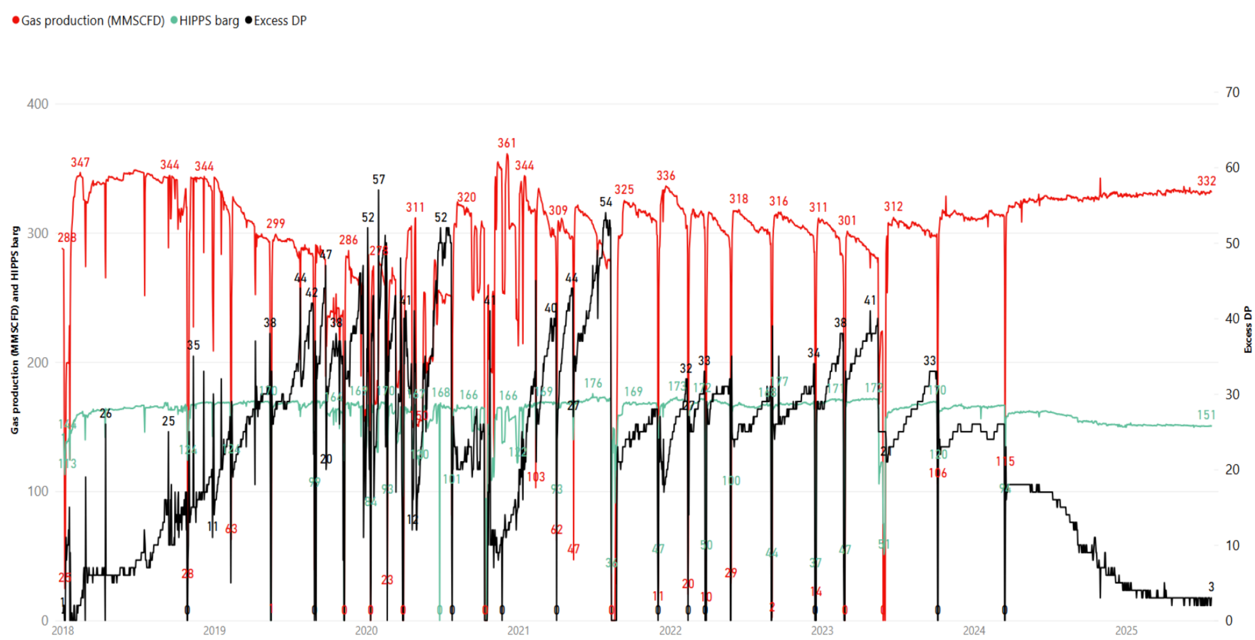

Figure 14—Atoll PL Performance Curve

Progressively optimizing the method. This innovative solution, engineered by the PHPC team, has played a key role in preserving pipeline integrity and ensuring uninterrupted gas and condensate delivery. Previously, interventions were required every 2 to 3 months to manage wax accumulation and pressure drop. With continuous improvements, the frequency has been reduced to just one intervention annually—unless performance degradation is observed. This advancement has significantly improved operational reliability and maximized

Conclusion

The Atoll field, characterized by its high-wax-content condensate with WAT of 37°C, provides a crucial case study in tackling complex flow assurance issues in deepwater gas production. Its experience highlights a vital approach to overcoming these challenges: implementing a pioneering technique known as "virtual pigging." This involves injecting Monoethylene glycol (MEG) through existing infrastructure, where the MEG flow generates shear force that helps dislodge accumulated wax and integrating it with complementary chemical treatments, which effectively mitigated severe wax and hydrate blockages that had been disrupting flow and reducing production. This innovative strategy not only restored and significantly increased gas and condensate production but also greatly reduced the frequency of required interventions. The Atoll approach serves as a valuable model for tackling similar issues in other deepwater oil and gas operations, particularly where conventional pigging methods are not feasible

Acknowledgement

We extend our sincere gratitude and acknowledge the dedicated efforts of all PHPC staff members involved in the development of the Virtual Pigging Technique. Your innovation and hard work have been essential in this achievement.

Abbreviation

Bbl	Barrel (of oil)
BBL/MMSCF	Barrels per Million Standard Cubic Feet
BP	British Petroleum
°C	Degrees Celsius
CPA	Cubic-plus-association (equation of state)
END	East Nile Delta
EPS	Early Production System
HIPPS	High Integrity Pressure Protection System
HPHT	High Pressure/High Temperature
JT	Joule-Thomson (effect)
MEG	Monoethylene Glycol
PL	Pipeline
PLEM	Pipeline End Manifold
TAR	Turnaround
WAT	Wax Appearance Temperature
XT	Christmas tree (subsea or surface wellhead assembly)

References

1. Billdal, Xin Chen. 2019. "Wax Deposition Study on 20″ Gas Condensate Trunk-line."
2. Centre, Bp Upstream Engineering. 2019. "Atoll - Flow Assurance Review of Chemical Soak options."

3. Chapman, Richard. 2016. "Wax Deposition Study for Atoll. BP Upstream engineering Centre. Rept. No. UE-2016-0246."
4. Harris, Lorraine. 2016. "Production Chemistry Fluid characterisation for Atoll. BP Upstream engineering Centre. rept. NO UE-2016-0223."
5. N. Haines. 2017. "Wax Management Strategy."
6. PhPC, Pharaonic Petroleum Company. 2016. "Atoll Development Project., Atoll EoS Benchmark, GE-ATZZZZ-PR-REP-0007_B1,."—. 2018. Wax Repair and Mitigation Strategy.